

SUMMER INTERNSHIP - 2026

AI Agents for Digital Capability Building in Tata Steel

WEEK-1

Think like a consultant, not a coder

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The Current L&D Environment

In large industrial companies such as Tata Steel, Tata Motors, Reliance Industries, or Larsen & Toubro, L&D usually stands for Learning and Development.

Typical Responsibilities (L&D)

- Identify skill gaps and design training programs to improve employee competencies and performance.
- Manage onboarding, technical, safety, compliance, and leadership development initiatives.
- Track training effectiveness and support career growth, succession planning, and workforce capability development.

Typical Programs

- Technical and role-specific skill development programs. (SAP/ERP training, AI workshops)
- Safety, compliance, and onboarding training. (Safety certifications)
- Supervisor and manager development programs, Graduate Engineer Trainee (GET), and digital/AI upskilling programs.

Some Current Industries L&D Setup

Topic	Tata Steel Ltd.	Mahindra Group
L&D Focus Areas	Focus on technical, managerial, leadership, safety, and digital capability development through Tata Steel Academy, TMDC, and SNTI.	Focus on continuous learning, leadership development, future skills, and workforce transformation through competency-based learning frameworks.
Learning Approach	Competency-based training, structured learning academies, blended learning (classroom + digital)	Competency frameworks, continuous learning culture, self-paced and role-based learning
AI / Analytics Adoption	Focus on Industry 4.0 skills, digital capability building, analytics-driven workforce development	Learning analytics, personalized learning recommendations, data-driven talent development
Digital Learning Platforms	Digie-Shala, Capability Development Portal, e-learning modules, classroom, lab-based and blended learning.	Enterprise Learning Experience Platform (LXP) promoting self-directed and continuous learning journeys.

Some Common L&D challenges in large manufacturing Setups

1. Skill Gap Visibility

Problem:

Management does not have a real-time view of employee skills.

Difficult to identify who lacks which competency.

Skill data is often spread across spreadsheets, certifications, and performance reviews.

Impact:

Wrong training assignments.

Delayed workforce planning.

Increased dependence on manual assessments.

AI Can be used for:

Skill inventory and competency mapping.

AI-generated skill gap analysis.

Role-to-skill matching dashboards.

2. Low Training Effectiveness

Problem:

Employees attend training but may not retain or apply knowledge.

Same training content is delivered to everyone regardless of background.

Impact:

Low ROI on training programs.

Productivity improvements are limited.

Repeated training requirements.

AI Can be used for:

Adaptive learning paths.

AI quizzes and assessments.

Personalized reinforcement recommendations.

Training effectiveness analytics.

3. Lack of Personalization

Problem:

A maintenance engineer, production engineer, and supervisor may receive similar training modules.
Learning journeys are not tailored to career goals or current skill levels.

Impact:

Reduced engagement.
Training fatigue.
Slower skill development.

AI Can be used for:

Personalized learning recommendations.
Role-based content generation.
Career-path-driven learning plans.

4. Knowledge Loss from Experienced Workforce

Problem:

Senior employees retire, resign, or transfer.
Critical tacit knowledge remains undocumented.
Examples: Troubleshooting specific equipment failures.
Operational best practices. And Plant-specific insights learned through experience.

Impact:

Repeated mistakes.
Longer onboarding time.
Loss of organizational expertise.

AI Can be used for:

Knowledge capture through interviews and documents.
RAG-based knowledge assistant.
Searchable expert knowledge repository.
AI chatbot trained on SOPs, manuals, and expert experiences.

The Primary Problems

1. Lack of a centralized system to assess, track, and visualize employee skills, competency gaps, and performance, limiting data-driven workforce development and training decisions.
2. Repetitive and unnecessary training for some employees. Lower engagement and training effectiveness and slower skill development.
3. Critical operational knowledge, troubleshooting expertise, and best practices are lost when experienced employees retire, resign, or move to different roles.
4. Organizations lack visibility into whether employees retain and apply knowledge gained from training programs.
5. Unified classroom training or expert sessions, lectures are mostly not taken seriously and often overlooked until forced upon them, and employees attend only occasionally.

Business Impact

- Inefficient allocation of training budgets and resources.
- Delays in workforce planning, succession planning, and deployment of talent to business-critical roles.
- Employees spend time on irrelevant or repetitive training.
- Lower engagement, learning effectiveness, and skill development speed.
- Reduced return on training investments and slower career progression.
- Loss of critical operational and domain expertise when employees retire or leave.
- Longer onboarding time for new employees and increased dependency on a few experts.
- Repeated mistakes, slower troubleshooting, and reduced operational efficiency.
- Training completion does not translate into improved job performance.
- Increased need for retraining, leading to higher training costs.
- Limited improvement in productivity, capability building, and business outcomes despite training investments.

Why AI ?

- AI can analyse training records, assessments, certifications, and performance data to automatically identify skill gaps.
- Provides real-time visibility into workforce capabilities at employee, team, and organizational levels. Helps managers make data-driven training and workforce planning decisions.
- AI can recommend personalized learning paths based on role, department, experience, and skill gaps. Continuously adapts recommendations as employees learn and progress.
- AI can capture and organize knowledge from SOPs, reports, manuals, and expert interactions. Makes critical expertise searchable and accessible through a conversational knowledge assistant.
- Identifies employees who need additional support or reinforcement.
- Preserves institutional knowledge and reduces dependency on individual experts.

Secondary Opportunity

Problem

The organization has limited visibility into whether training programs actually improve employee skills and job performance.

Gap In System

Training completion is tracked, but learning outcomes and workplace application are not consistently measured.
Limited feedback loops exist to improve training content and delivery.

Impact on Business

Training investments may not deliver expected returns.
Employees may require repeated training on the same topics.
Difficult to identify which programs are effective and which need improvement.

Why AI?

AI can analyse assessment scores, training participation, performance data, and feedback to measure learning impact.
It can identify at-risk learners, recommend reinforcement, and provide insights on training ROI.